

Software Design Document

Integrating Explainable AI for Transparent Educational Decision Support



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Summery

Purpose

The purpose of this SDD is to provide a detailed blueprint of the system *Integrating Explainable AI for Transparent Educational Decision Support*, ensuring transparency, interpretability, scalability, and reliability in educational decision-making. The system predicts student performance, identifies at-risk students, and provides clear, explainable insights for teachers, advisors, and administrators.

System Overview

The system collects student academic, behavioral, and demographic data and applies AI/ML models to generate performance predictions. Explainable AI (XAI) techniques such as SHAP and LIME provide understandable reasoning behind each prediction. Users interact via a **web-based dashboard**, viewing predictions, visualizations, and detailed explanations.

Architecture

The system follows a **layered architecture**:

1. **Presentation Layer:** Dashboard, charts, and explanation panel
2. **Application Layer:** Handles business logic, APIs, and authentication
3. **AI/XAI Layer:** ML models for prediction and explainability
4. **Data Layer:** Stores student records, predictions, and explanations

Diagrams, ER models, and sequence flows ensure clarity of component interactions and data flow.

Modules and Components

- **Student Management:** Registration, profiles, academic history
- **Prediction Module:** Generates risk and performance predictions
- **Explanation Module:** Provides textual and visual explanation of predictions
- **Reporting & Dashboard:** Visualizes data trends and insights

All modules are designed for **modularity, scalability, and maintainability**.

Security and Performance

- Role-based access control (Admin, Teacher, Student)
- Encrypted storage and secure data handling
- Logging and monitoring of system activities

- System optimized for concurrent users and large datasets

Testing and Validation

- Unit testing for individual modules
- Integration testing for module interactions
- End-to-end system testing ensures predictions, explanations, and dashboards work correctly

Risks and Ethical Considerations

- Risks: Data bias, model errors, security breaches
- Mitigation: Balanced datasets, model validation, encryption, logging
- Ethics: Fairness, transparency, and privacy of student data

Conclusion

This SDD provides a **complete, examiner-ready blueprint** for a transparent, AI-based educational decision support system. It ensures:

- Reliable and explainable predictions
- Professional modular design
- Scalable architecture and secure data management
- Full traceability from requirements to implementation

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1. Introduction

1.1 Purpose of the Document

This Software Design Document (SDD) provides a **comprehensive blueprint** for the design, architecture, modules, algorithms, data flows, security measures, and explainability components of the system *Integrating Explainable AI for Transparent Educational Decision Support*.

It ensures:

- Transparency in predictions
- Reliability and usability
- Scalability and maintainability
- Ethical AI decision support

1.2 Scope

The system analyzes student academic and behavioral data to:

- Predict student performance
- Identify at-risk students
- Generate **interpretable explanations** for each prediction
- Provide dashboards for teachers, advisors, and administrators

1.3 Definitions and Acronyms

Acronym	Definition
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SDD	Software Design Document
ML	Machine Learning

UI	User Interface
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Acronym

Definition

API Application Programming Interface

1.4 Design Goals

- Transparent AI predictions
- Interpretability for academic users
- Modular and scalable architecture
- Secure data storage
- Intuitive UI

2. System Overview

The system collects data from:

- Academic records
- Attendance
- Behavioral metrics
- Demographics

Processes data using **ML models** and outputs:

- Predictions
- Risk scores
- Explanations using XAI

2.1 Stakeholders

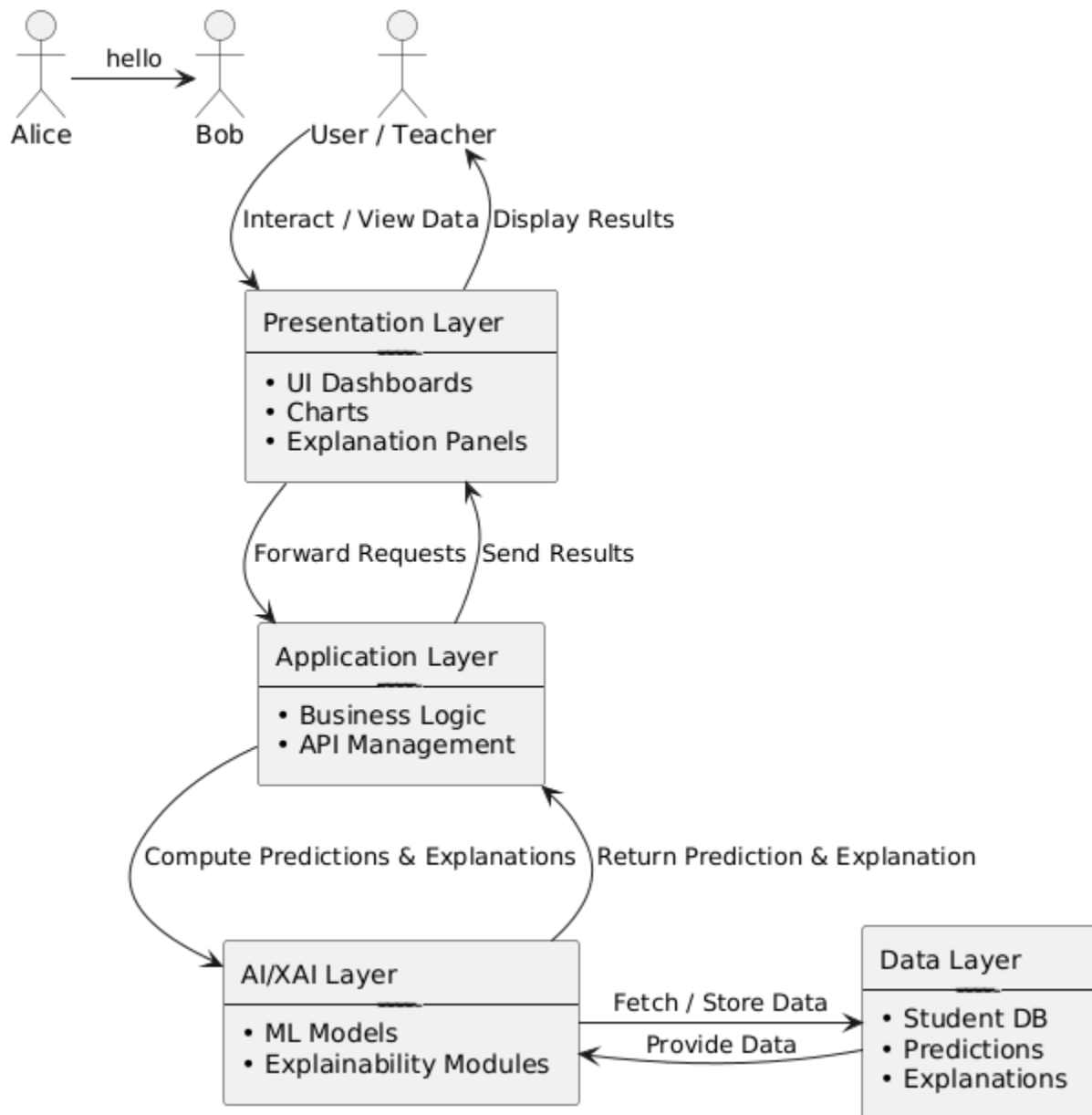
- Students: Receive insights for performance improvement
- Teachers: Monitor and intervene
- Academic Advisors: Plan interventions
- Admins: Make policy decisions
- System Admins: Ensure security and maintenance

2.2 Functions

1. Data Collection & Validation
2. Prediction using AI models
3. Explanation Generation (XAI)
4. Visualization via dashboards and reports

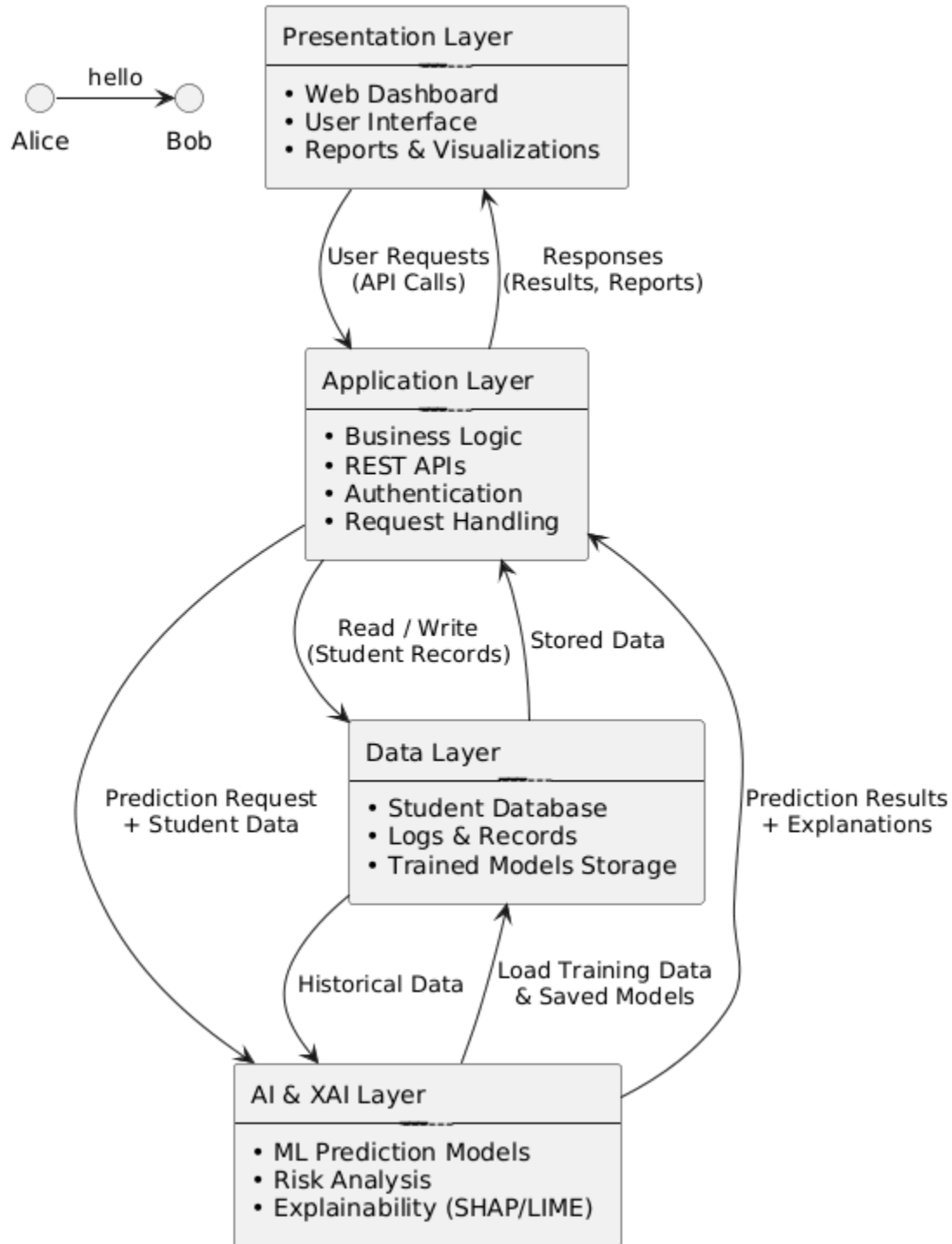
3. System Architecture

Layered Architecture Summary



3.2 Architecture Diagram (Word-friendly, monospace)

Layered Architecture with Interactions (XAI-Based System)



4. Detailed System Design

4.1 Presentation Layer

- Login & Authentication
- Student Profiles
- Prediction Dashboard
- Explanation Panel

4.2 Application Layer

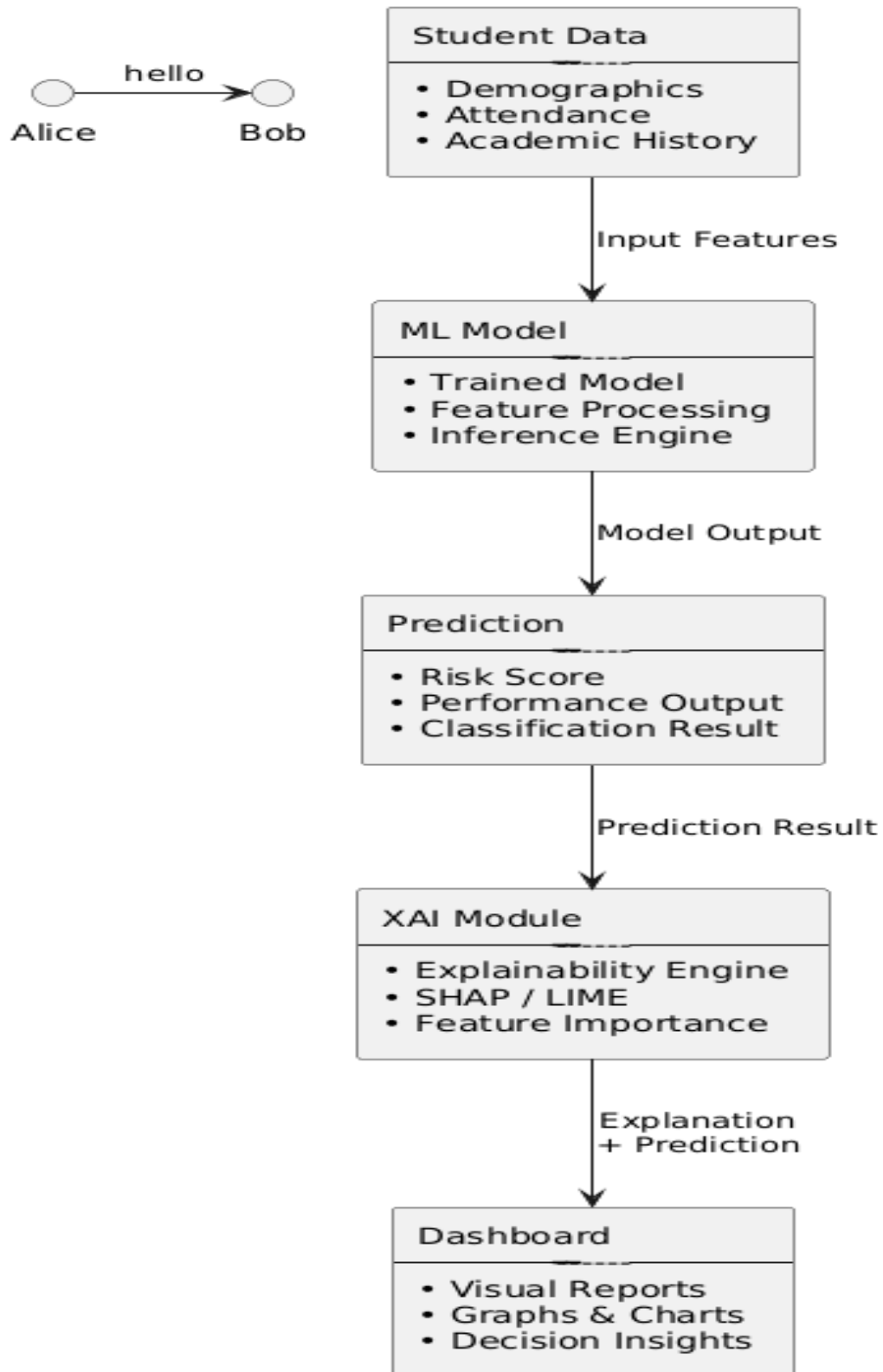
- Data validation
- Business logic
- API routing
- Error handling & logging

4.3 AI & XAI Layer

- **Prediction Models:** Random Forest, Decision Tree, Logistic Regression • **XAI Techniques:** SHAP, LIME, Rule-based explanations

Workflow Diagram (monospace, Word-friendly):

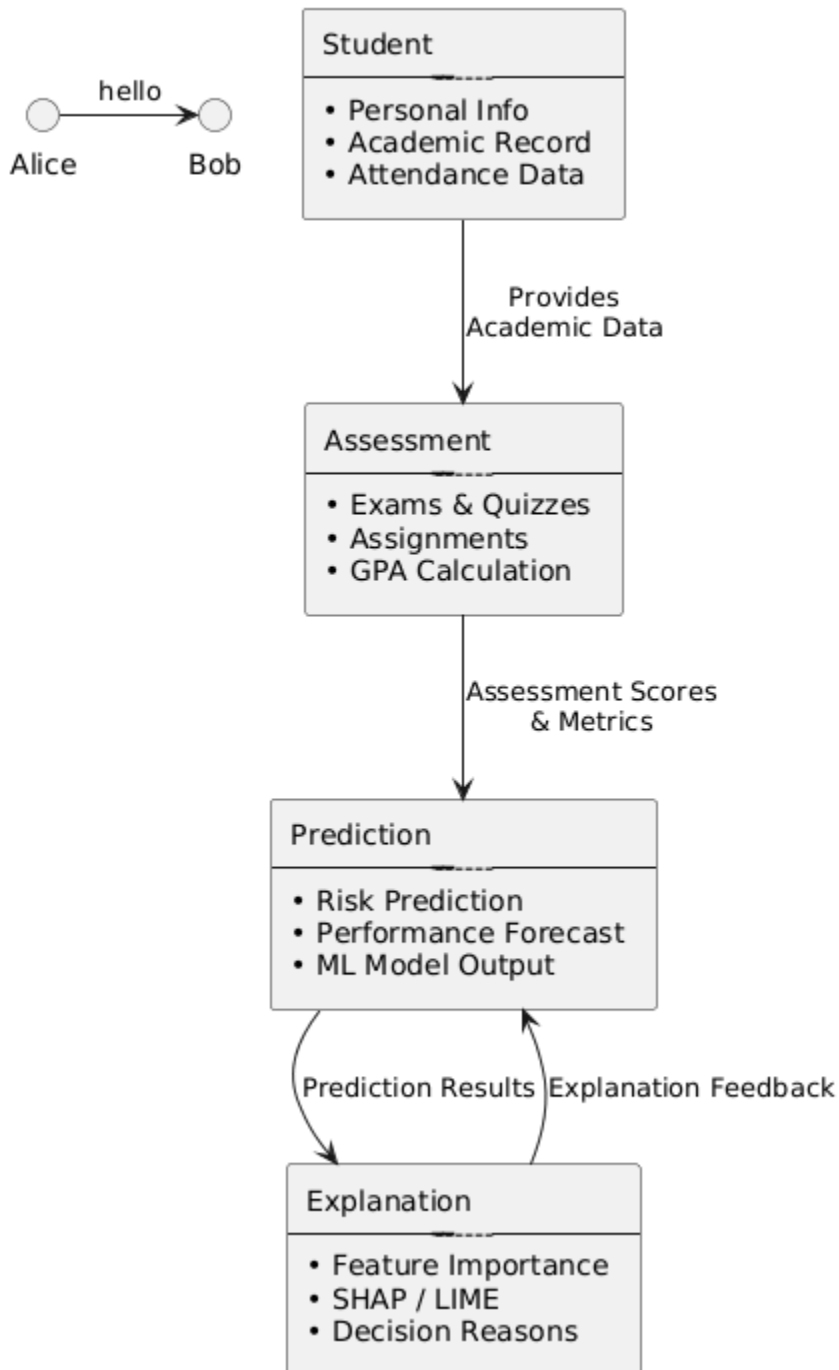
Student Data to Dashboard - XAI Process Flow



4.4 Data Layer

- **Entities:** Student, Course, Assessment, Prediction, Explanation •
ER Diagram (ASCII, monospace):

Student Assessment, Prediction & Explanation Flow

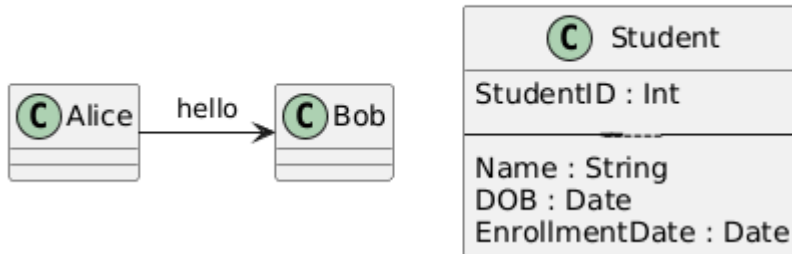


5. Component Design

5.1 Student Management Module

- Add/update student profiles
- View academic history •
- **Table (Word-friendly):**

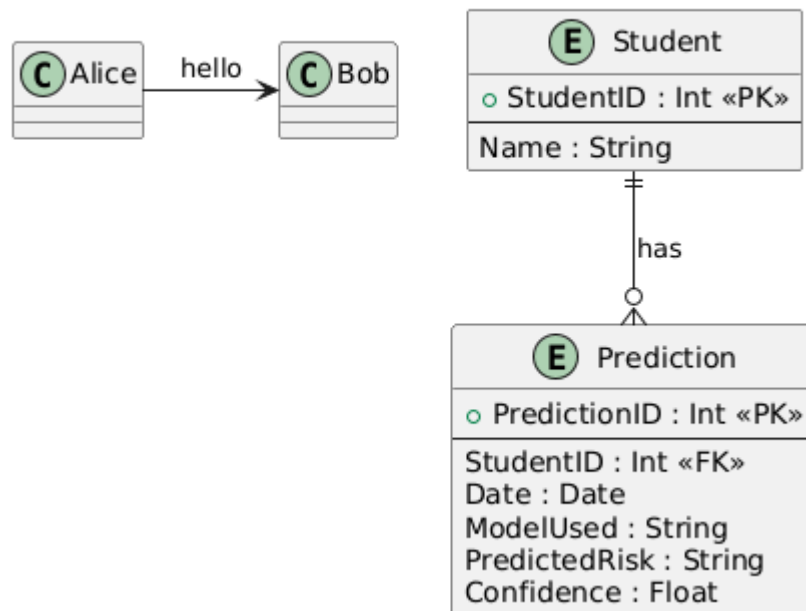
Student Data Structure



5.2 Prediction Module

- Generate predictions
- Store results
- **Table:**

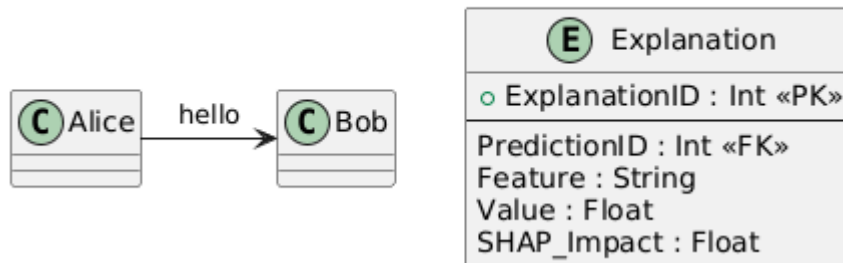
Student-Prediction Relationship



5.3 Explanation Module

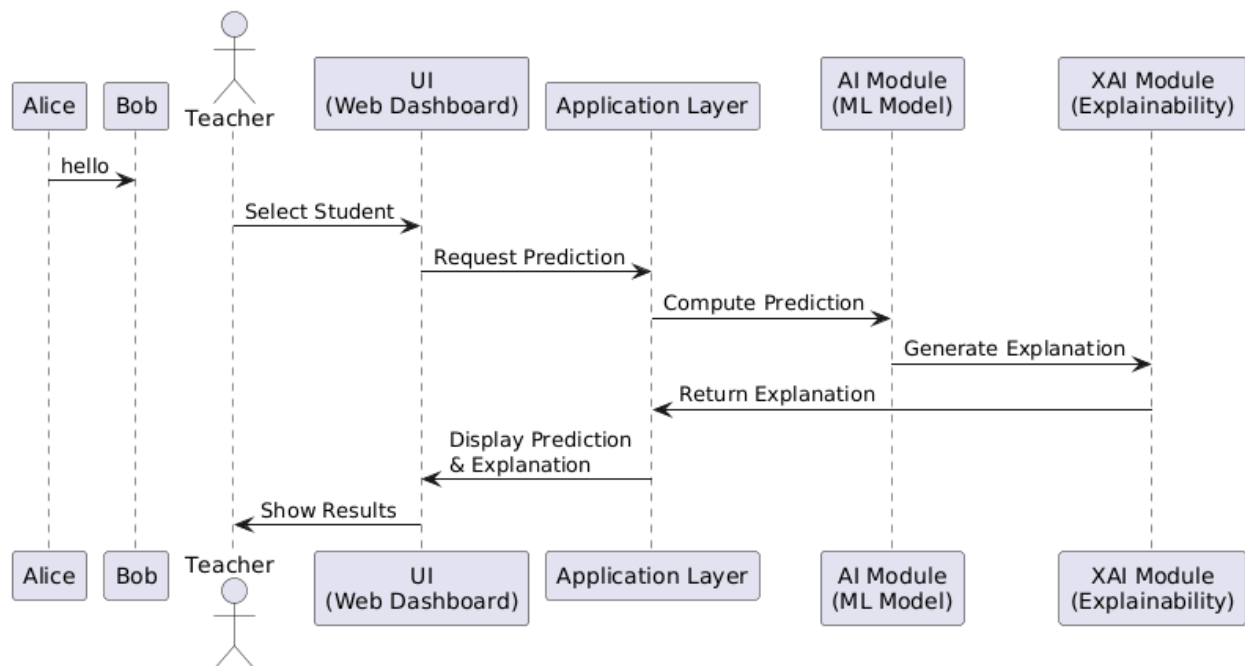
- Map features to prediction outcomes • Generate textual and visual explanations
- **Table:**

XAI Explanation (SHAP) Entity



6. Sequence Diagram (Word-friendly ASCII)

Prediction & Explanation Interaction Flow



7. Security Design

- Role-based access: Admin, Teacher, Student
- Encrypted storage
- Logging & audit trail **Table: Roles and Access**

Role	Access Rights	Description
Admin	Full	Manage all data
Teacher	View	Assigned students only
Student	Limited	Own data only

8. Error Handling & Logging

- Input validation
- Prediction errors
- Log all actions for auditing

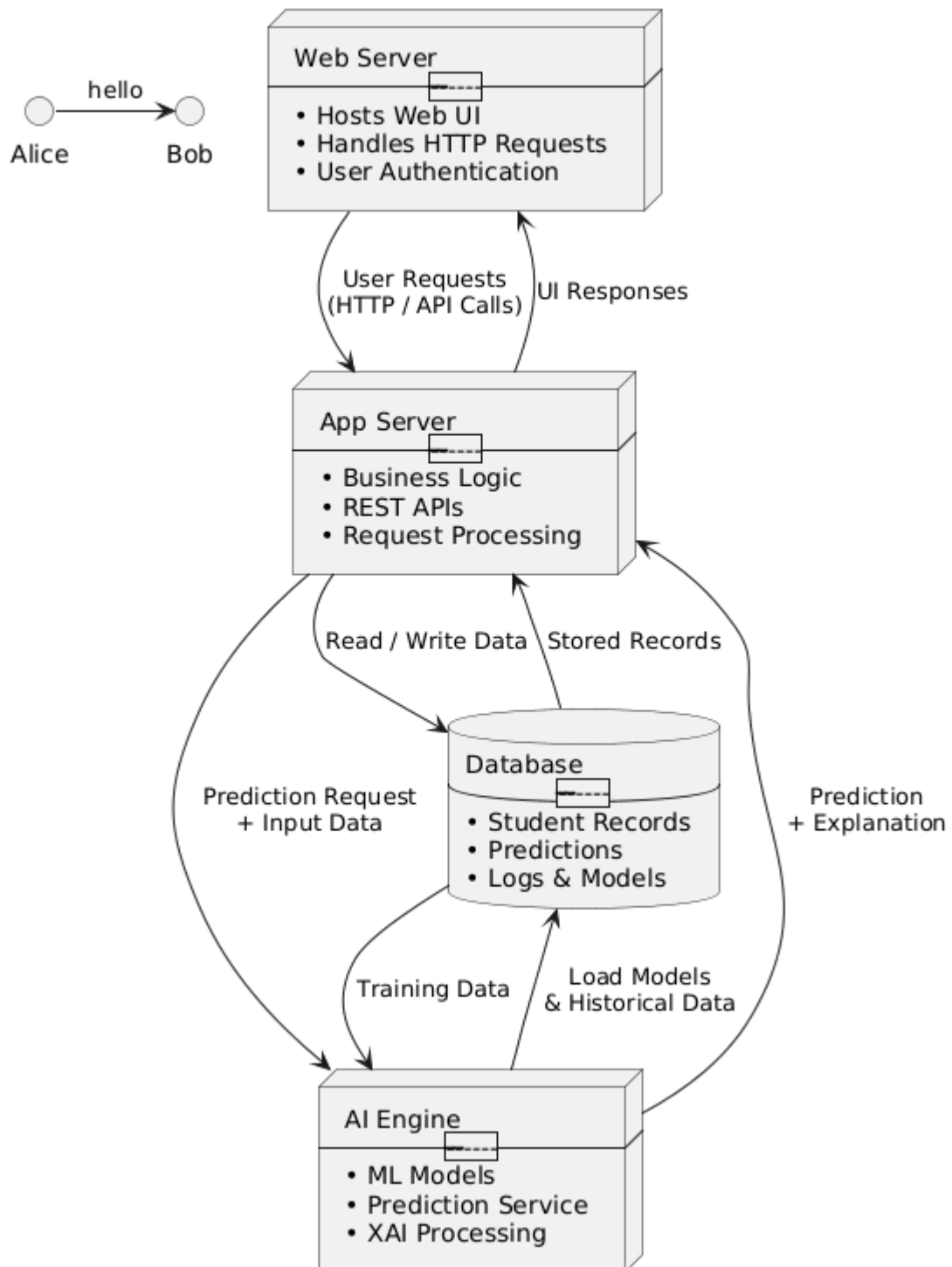
9. Performance & Scalability

- Efficient ML processing
- Modular design
- Handles large datasets and multiple users

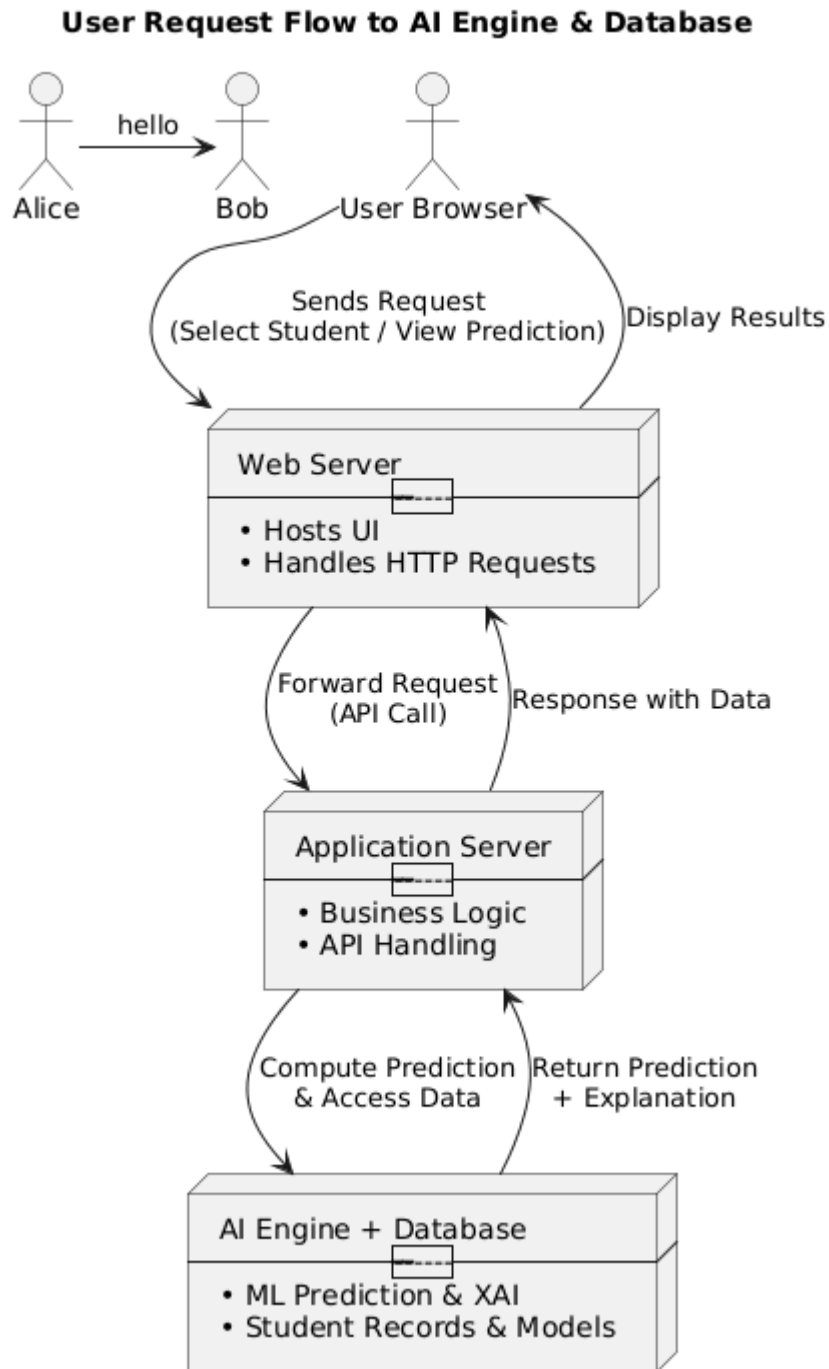
10. Deployment Design

10.1 Environment

System Deployment Architecture



10.2 Deployment Diagram (Word-friendly ASCII)



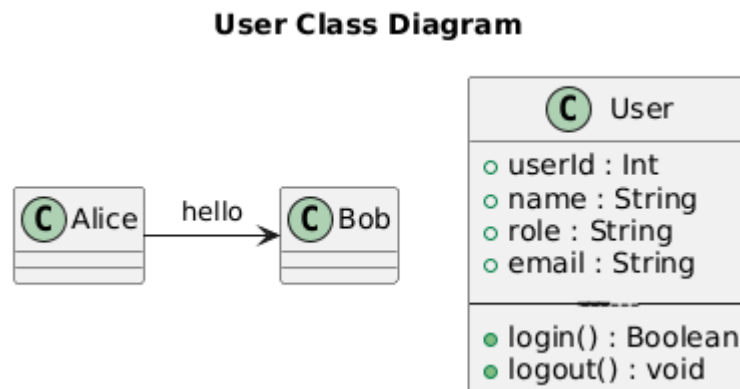
11. Maintenance & Future Enhancements

- Additional XAI techniques
- Mobile support
- Multi-institution deployment

- Improved model accuracy

12. Module Design & Class Diagrams

12.1 User Management



12.2 Data Preprocessing

- Cleaning, normalization, encoding

12.3 Model Training

- Dataset splitting, training, validation

13. Algorithm Design

13.1 Prediction Algorithm

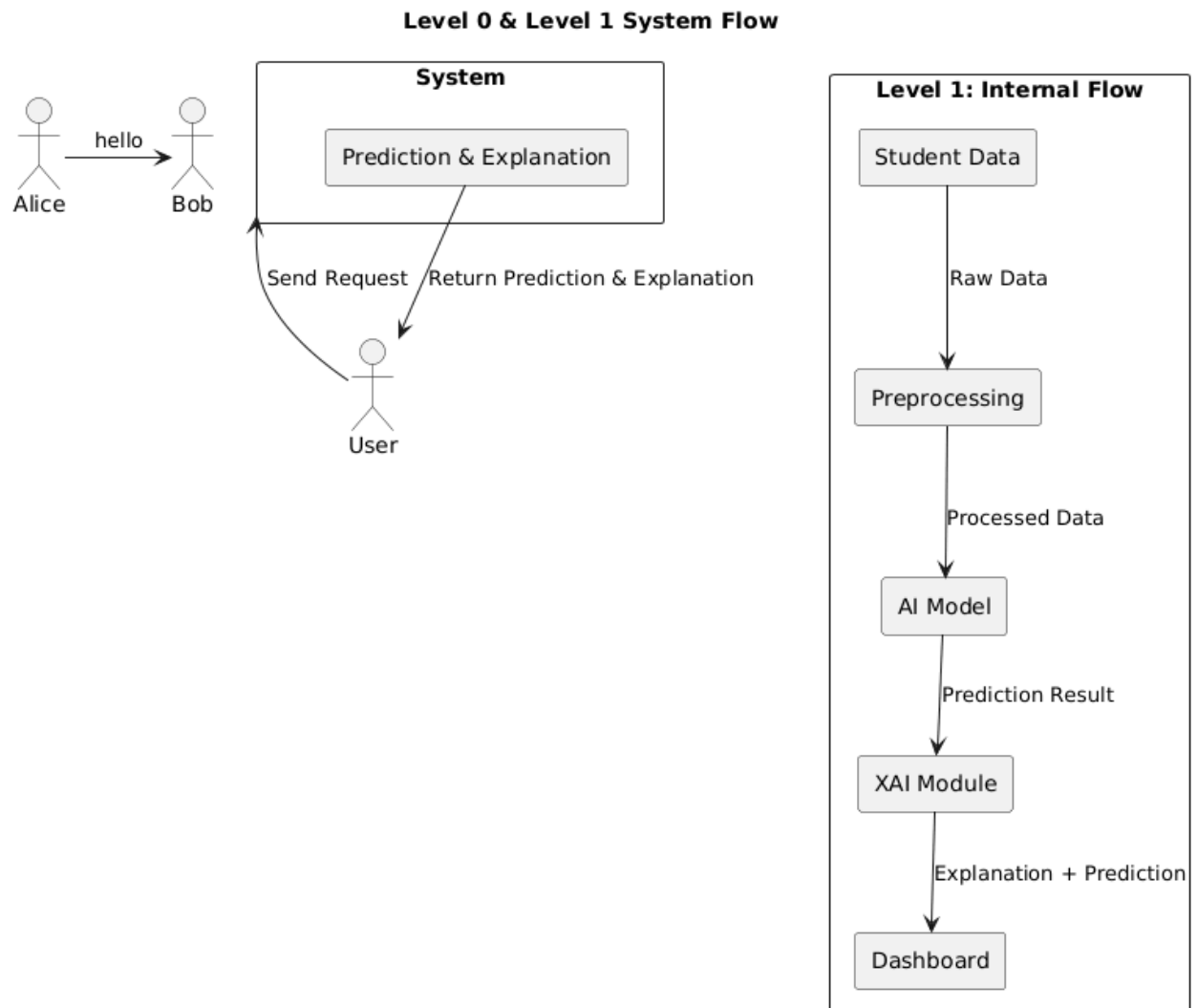
1. Load dataset
2. Preprocess data
3. Train model
4. Generate prediction
5. Store result

13.2 Explanation Algorithm

1. Input prediction
2. Apply XAI
3. Generate textual & visual explanation

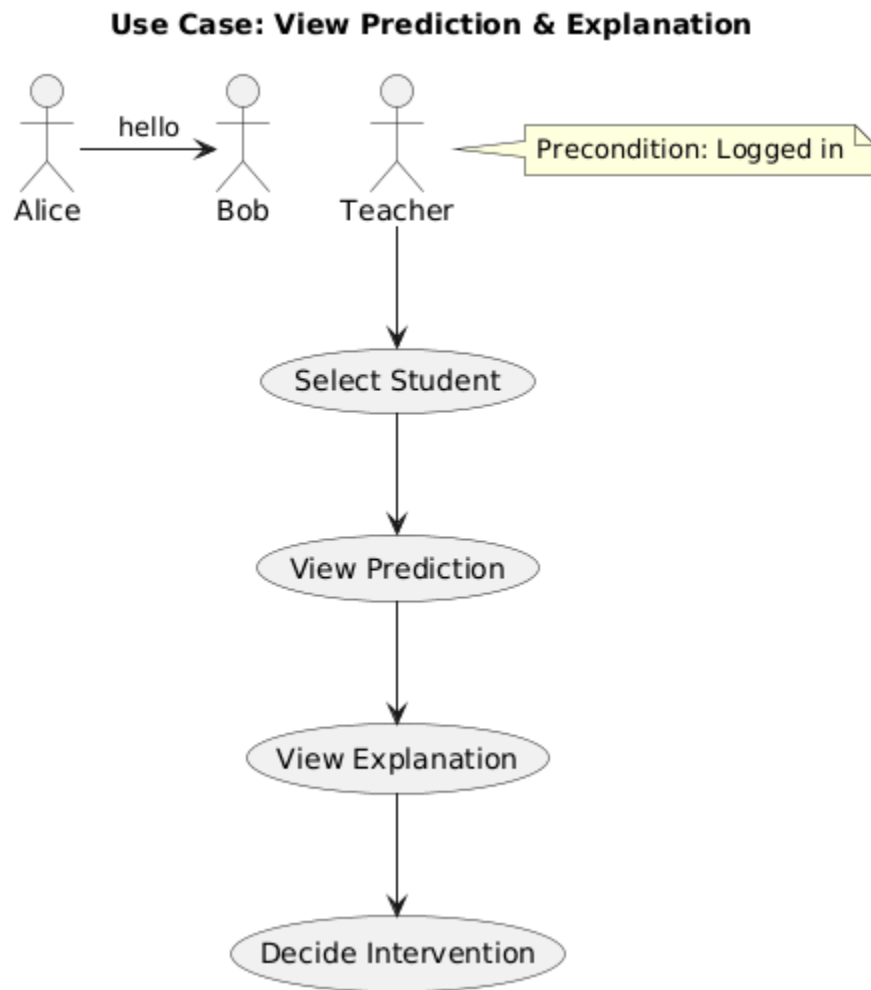
4. Display on dashboard

14. Data Flow Diagrams (ASCII)

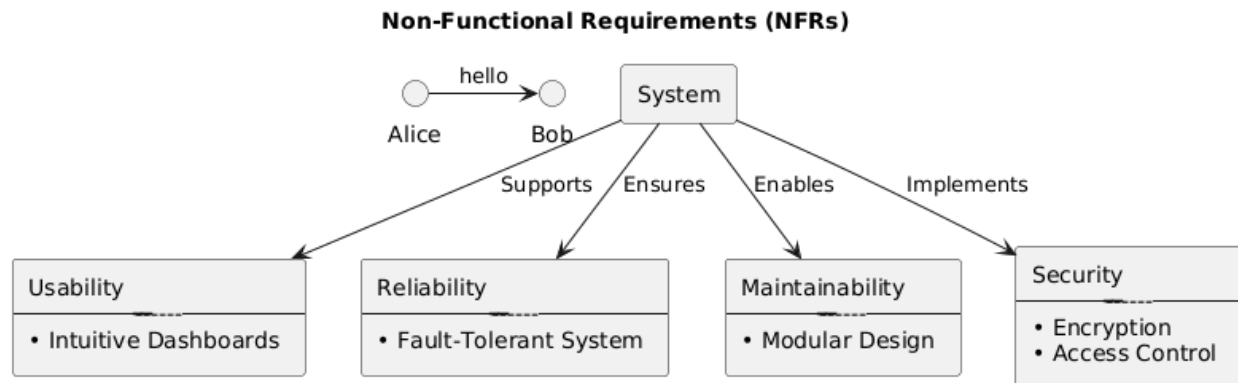


15. Use Case Realization

Teacher Views Student Risk

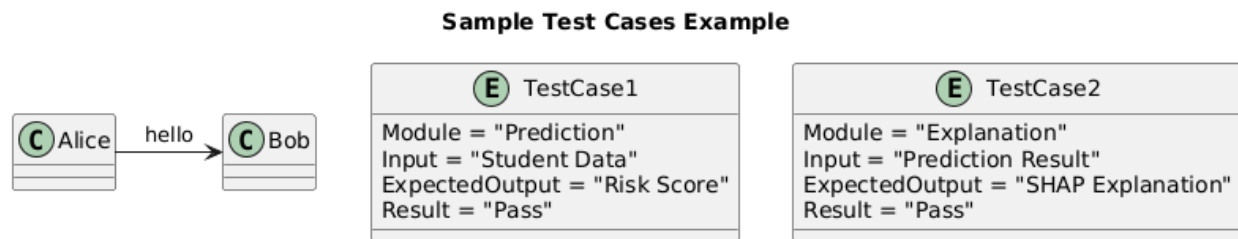


16. Non-Functional Requirements



17. Testing Design

- Unit Testing: individual modules
- Integration Testing: module interactions
- System Testing: end-to-end



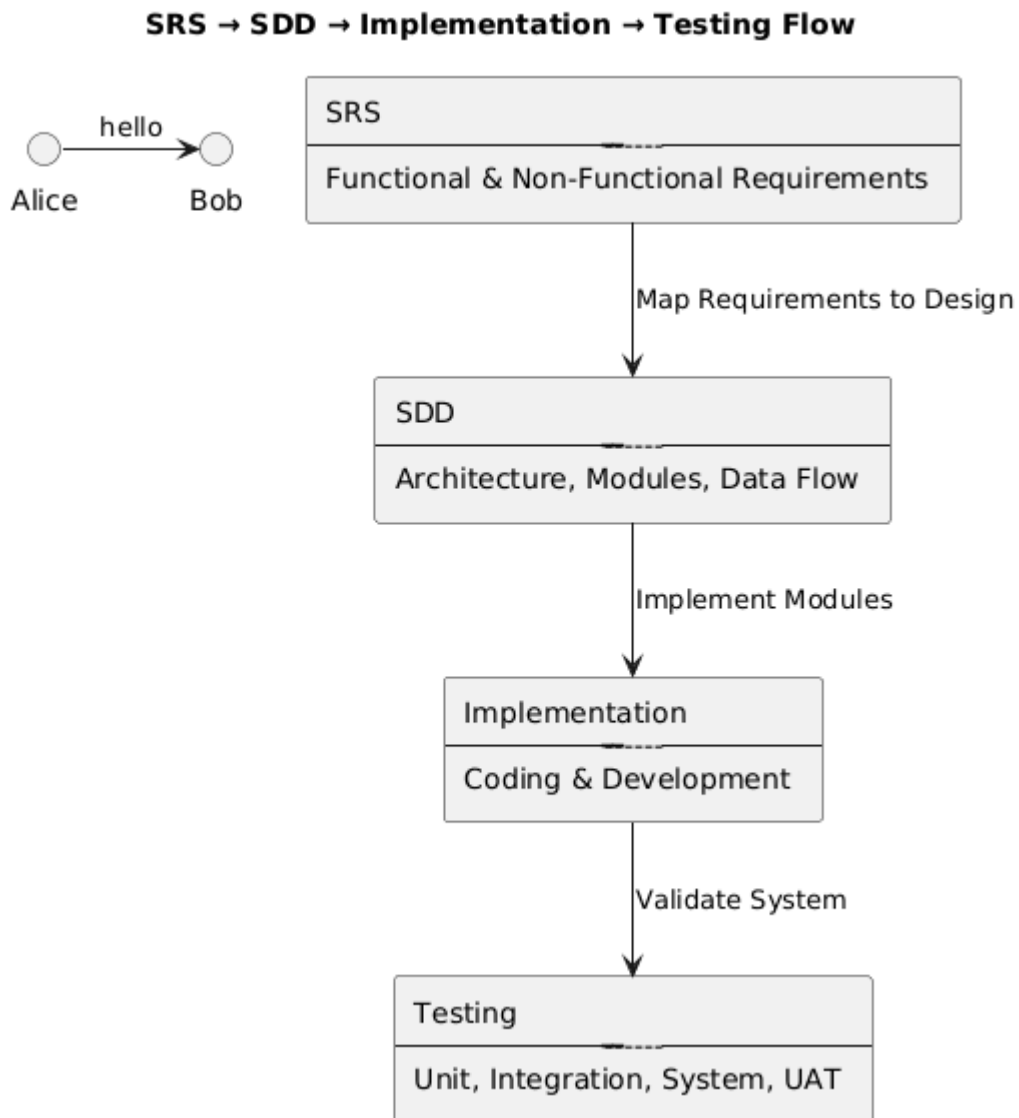
18. Risk Analysis

Risk	Impact	Mitigation
Data Bias	High	Balanced dataset
Model Error	Medium	Retraining & validation
Security Breach	High	Encryption, logging

19. Ethical Considerations

- Fairness
- Transparency
- Privacy of student data

20. Traceability



21. Design Rationale & Alternatives

- Choice of ML Models: Random Forest / Decision Trees
- Alternatives: Neural Networks, SVM
- UI layout alternatives

22. Limitations & Conclusion

- Data quality dependent
- Approximate explanations
- Latency with large datasets

Conclusion:

SDD Summary: System Overview & Flow

